Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 21 AV Summary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 2 Class 21 AV Summary

1 UE26ME142A Unit 2 Class 21 AV Summary

Study focus: UE26ME142A_Unit 2_Class 21_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000186 · EV0000187

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Vengadesh from Civil Engineering under the Department of Science and Humanities, PS University.
- This is class number thirty-one.
- We are in the process of understanding.
- Chapter 4, that is trusses under unit 2.
- In the last class I had explained about what is a truss, what is trust made up of, what are the different types of trusses that we have, what is plain truss, what is.
- The load that can be applied on trusses as well as what is tension forces acting on the members and what is compressive forces acting on the members.
- And also I talked about what are the different methods of determining the forces in the members.
- which is either in going to be in the compression or at tension.
- which explains what are the different methods of determining the forces in the members.
- The first method is method of joint.
- There's one more method which is called as method of section, but in the syllabus prescribed for you, we have only method of joint, so we'll concentrate only on method of joints.
- Since there is no horizontal force acting on the truss, there is no horizontal reaction acting at A.
- Now coming to method of joints, the first step that is involved in method of joint is to identify a joint which has got only two unknowns.

- None of the joints have got only two unknowns unless and until we determine the support reaction R1 and R2.
- If we determine the support reaction R1 and R2, there will be two joints that is joint A and joint D which will have only two unknowns.
- That is in member AF and member AB as well as if you take joint D it will be member DC and DF.
- However, here we have considered a joint A, that is this joint A.
- is the external force, which is the reaction force, that is R1, and then there are two members, AF and AB.
- I am assuming the force in the member A B as tension.
- This is external force, this is internal force.
- Next the force in A the force in AB is considered to be moving away from the joint, which is under tension.
- Whereas the internal forces will be acting away from each other.
- This is the external force is acting towards there will be one more here.
- Internal forces is acting away from each other.
- So that means it will be towards this joint.
- AB that is this angle.
- I can resolve the force in the member AF along vertical as well as along horizontal and then use the equations of equilibrium to determine the forces in the member AF and AB.
- There are two equations of equilibrium, so we can determine these two forces.
- Next after taking this joint A, next we have to choose the other joint which has got only two unknowns.
- Again, we can resolve the force in the member AF along horizontal and vertical, use the equations of equilibrium to find out the forces in F E and F.
- Once I determine these two members, now I can take joint C, I can determine the forces in these two members.

Example or application from the lecture

- Now to explain the method of joints I must take a simple example as plane trust ABCDEF At A we have a pin connection, at D we have a roller connection and there is a force applied at B which is here which is also vertically downwards.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then I can take joint E where I will be knowing all these three members.

Key details

- There will be only one non-on which is E D which I can solve.
- Then if I take joint D, now we know the force in the member E D as well as the force in the member C D.
- And if I had already determined R2, I would also be knowing R2.

- But what we could do here is we can take R2 as an unknown.
- Determine R2 by using the forces that we have got from the member ED and C D, and check whether we get the same answer, whatever we would have got earlier.
- in the members.
- Now in the entire engineering mechanics, whatever that is included in the syllabus, only for trusses we would be able to say whether the forces that we have got in the members are correct or not because we have a chuck in this case.
- That is under tension.
- Then if you take the member F B, again the forces are acting towards each other, that is under compression.
- Then if you take FE, the forces are acting away from each other is under tension.
- This is the method that we should use to get the forces in the members.
- I hope all of you have understood.
- The first thing is you have a member A B C D okay like this, it's a square one.
- And then we also have the member.
- B C and AB.
- This is just overlapping each other and each of them can move individually without the joint considering them.
- Now if because of the external forces, if the body is struck like this,.
- And also if you observe at each of the joint there are more than three members.
- Here these diagonal members are capable of supporting only tension.
- It cannot support compression.
- This member AB and D are cables which can take tension which cannot take compression cable or row.
- You can easily identify that this diagonal T C is getting compressed, whereas this diagonal AB is getting elongated.
- So because of which I can say that.
- The force in the member A B will be under tension, whereas the force in the member D C will be under compression.
- Then I can take this joint since I have no force in this, I can find out the force in these two, and similarly I can go out and join the other joints also.
- Then now coming to other special cases if there are only three members of the joint and if two members are.

Example or application from the lecture

- All the joints are given here, and then finally we will find out the forces in all the members, and once we get the forces in all the members, then we will represent those forces on the truss like this.
- For example, if you take AF, since the forces are acting away from each other, they are internal forces.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 10:00–14:59

Collinear, okay, the forces in these two members are going to be equal because of which the force in this member when I resolve it, if I take some of all the forces along this, the force in this member will be simply zero.

Key details

- Only two members of the joint and they are acting like this, then if you take any axis x and x dash, the forces in those two members will be zero.
- Then if you have a crossing of a members, since this member is collinear, the member in these two four.
- forces in these two members will be same similarly the forces in these two members is going to be same it is not essential that they have to be at retire decrease only so if the members are collinear if they are in a straight line then the forces in those two members is going to be same okay.
- Problem 4 bar 5 is what is taken here.
- And you can also look at those joints.
- At joint A, there are only two members since we know the load at A already, so there are only two unknown forces that is acting.
- And the force acting in this member is 4 kilometers.
- I think that is missing here.
- Take that force acting at joint A as 4 kilometers.
- Angle BAE which is going to be this angle.
- Why am I doing this?
- Because it is easier to say initially that the forces are moving away from the joint.
- And once I get the values of this, if it if my assumption is correct, I get a positive answer.
- If my assumption is wrong, I'll get a negative answer, which indicates that the force in that member is a compressive force.
- This one, this force in this member AF is acting away from A.
- I get the force in this member F A B.
- This is forty five, so sine forty five which is acting upwards.
- And the force, the external force which is acting downwards is 4 kilonewtons.
- Next, I'll take the other equation of the equilibrium that is summation of all the forces, algebraic sum of all the forces along X equals 0.
- AB cos 45 which is acting towards the right so positive.
- We will keep the sign positive or negative as it is.
- When we are substituting somewhere else for the value of this member forces A.
- I will take the sign also into consideration as it is as negative.
- Or if it is positive, I will take it as positive.
- Because I am assuming the forces in all the members as tension itself that is moving away from each other, it will help me out very much.
- Next I will take joint E.
- Then if I take sigma f x is equal to zero.
- we get the force in the member E D to be same as the force in the member.

- AE or EA, both of them are same, which is also going to be equal to when I substitute the value of minus 4, I get this to be minus 4.
- Next I will move on to joint B.

Example or application from the lecture

- This is the trust that is given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 15:00–19:59

FAB with a negative sign.

Key details

- Next, I'll take summation of all the forces along X is equal to 0.
- If I solve the joint E I will get E B.
- Then if I solve for joint B I will get the force in member B D.
- Okay now using the special conditions I can initially say that the force in the member BE is equal to zero because there is no force acting at E.
- And these two member that is A and E D are collinear, the forces in those two members are going to be equal.
- We have got the force in the member BE as zero, okay because there is no force acting at E and B D as minus five point six kilometer which indicates that it is under compression.
- I hope all of you have understood this.
- I've considered four.

Example or application from the lecture

- Let's move on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 21_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.

4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.
9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

C Revision and retrieval practice